

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV **COURSE CODE: DSA0202**

COURSE OUTCOME COVERED

CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

Assessment Tool Weightage: 30%

QUESTIONS: 2 Total Marks:30 Annexure A

Industry Problem Solving Task

Code of Conduct:

*I JAASIEL J S (192524046) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Industry Problem	20%	Comprehensive understanding of real world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	

Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Annexure A

Industry Problem Solving Task

1: Automated Defect Detection in Manufacturing

A precision engineering company wants to automatically detect defects in mechanical components using computer vision. Parts vary slightly in shape due to manufacturing tolerances.

Questions:

1. How can PCA-based shape priors and contour-based representations be used to model standard component shapes?

ANS: Contour-based representation extracts the boundary of each standard component from its image. The contour points are aligned and normalized, and PCA is applied to the training shapes to obtain a mean shape and principal modes of shape variation. These modes capture the normal changes caused by manufacturing tolerances. A new component is compared with this PCA-based shape prior; small deviations are treated as normal variation, while large deviations can indicate a defect.

2. Propose a pipeline using machine learning (SVM, GMM) or deep learning (CNN) to classify defective vs. non-defective parts.

ANS: A suitable pipeline is:

Image acquisition → preprocessing → contour/feature extraction → feature representation → classifier training → defect classification

For machine learning, SVM can classify extracted shape features into defective and non-defective classes, while GMM can model the feature distributions of the components. For larger labeled datasets, a CNN can learn features directly from component images and classify them as defective or non-defective.

3. Discuss methods to handle noise, partial occlusions, and varying illumination in industrial images.

ANS:

- **Noise:** Apply preprocessing such as filtering and denoising before shape analysis.
- **Partial occlusions:** Use contour/shape information from the visible regions and robust feature representations so that missing portions do not completely prevent recognition.
- **Varying illumination:** Use image normalization and illumination correction so that the extracted features are less affected by brightness changes.
- These preprocessing and robust-representation steps improve the reliability of defect classification.

2: Autonomous Vehicle Obstacle Recognition

A self-driving car company needs to detect rare or unexpected obstacles on urban roads. The labeled data for these obstacles is limited.

Questions:

1. Explain how data augmentation using GANs can help generate synthetic images for training.

ANS: GANs can generate realistic synthetic images of rare or unexpected obstacles such as unusual objects on urban roads. These generated images can increase the diversity and size of the training dataset when real labeled examples are limited. The synthetic images can include variations in object appearance, position, scale, and scene conditions. After quality checking, they can be combined with real images to improve the obstacle-detection model's ability to recognize rare objects.

2. Compare the effectiveness of CNNs versus Vision Transformers for detecting small or sparse objects in high-resolution images.

ANS: CNNs are effective at detecting local patterns such as edges, textures, and object features and are generally more computationally efficient. They are therefore suitable for real-time embedded systems.

Vision Transformers (ViTs) use attention mechanisms to capture relationships across the image and can model long-range context effectively. This can help when detecting small or sparse obstacles whose surrounding context is important, especially in high-resolution images. However, they generally require more computational resources and larger training datasets.

Therefore, CNNs are preferable when computational efficiency and real-time processing are critical, while Vision Transformers can provide stronger global-

context modeling when sufficient computation and training data are available.

3. Suggest a real-time inference pipeline for obstacle detection, considering computational constraints in embedded systems.

ANS: A suitable pipeline is:

Camera input → Image preprocessing → Lightweight CNN-based detector → Non-maximum suppression → Obstacle classification/localization → Confidence filtering → Decision/output

To satisfy embedded-system constraints, use a lightweight detection model, reduced input resolution where appropriate, optimized inference such as model quantization, and hardware acceleration. Only high-confidence detections should be passed to the vehicle's decision-making system.

This provides a balance between detection accuracy, latency, and computational efficiency for real-time obstacle recognition.